

# Carles Trullàs

## Data Scientist & ML Engineer

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## Profile

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Data Scientist with a background in Mathematics, providing a very solid foundation in modeling, programming, and analytical thinking. Ranked first in class throughout high school, university, and graduate studies, showcasing strong motivation and achievement.

## Education

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| <b>MSc</b> | <b>Universitat Oberta de Catalunya</b> , Data Science                                                                                                                                                 | Sept 2023 – Jul 2026  |
|            | <ul style="list-style-type: none"><li>Specialized in Machine Learning, Deep Learning, and AI Agentic Frameworks to build autonomous systems.</li><li>Honorable Mention in Machine Learning.</li></ul> |                       |
| <b>MSc</b> | <b>Polytechnic University of Barcelona</b> , Advanced Mathematics and Mathematical Engineering                                                                                                        | Sept 2022 – June 2023 |
|            | <ul style="list-style-type: none"><li>The final qualification of the master's degree is 9.8 (best of the promotion).</li><li>Awarded with 6 Honorable Mentions, including Master's Thesis.</li></ul>  |                       |
| <b>BSc</b> | <b>Autonomous University of Barcelona</b> , Mathematics                                                                                                                                               | Sept 2018 – June 2022 |
|            | <ul style="list-style-type: none"><li><i>Extraordinary award</i> for the best academic record of the promotion.</li><li>Awarded with 10 Honorable Mentions, including BSc final Thesis.</li></ul>     |                       |

## Experience

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| <b>NTT Data. Business Advances Analytics</b> , Data Scientist & ML Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Barcelona, Spain<br>Mar 2026 – Present     |
| <ul style="list-style-type: none"><li>Built a data-driven anomaly detection solution that identifies abnormal ordering patterns using ML-based demand forecasts, enabling early intervention before operational impact.</li></ul>                                                                                                                                                                                                                                                                                                                      |                                            |
| <b>UVE Solutions</b> , Data Scientist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Manresa, Spain<br>Sept 2024 – Mar 2026     |
| <ul style="list-style-type: none"><li>Designed and deployed an AI-powered search engine capable of finding Points of Sale with photos and reviews most similar to a given image or short text input.</li><li>Delivered client-focused analytics to identify high-potential points of sale, supporting international companies in optimizing their commercial strategies.</li><li>Collected data from multiple sources via scraping and performed ETL processes to produce clean, structured datasets ready for analysis and decision-making.</li></ul> |                                            |
| <b>Autonomous University of Barcelona</b> , Assistant Lecturer (Faculty of Mathematics)                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Bellaterra, Spain<br>Sept 2023 – Aug 2024  |
| <ul style="list-style-type: none"><li>Taught courses in Statistics, Differential Equations, Linear Algebra, and Calculus.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |
| <b>Center of Mathematical Research (CRM)</b> , Scholarship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Bellaterra, Spain<br>Jan 2023 – Aug 2023   |
| <ul style="list-style-type: none"><li>I received a scholarship to work on my master's thesis in advanced mathematics.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                            |
| <b>Group on Dynamical Systems, UAB (GSDUAB)</b> , Research Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Bellaterra, Spain<br>July 2021 – Sept 2021 |
| <ul style="list-style-type: none"><li>Intern at the <i>Group on Dynamical Systems of UAB</i>, which belongs to the Mathematics Department at the Autonomous University of Barcelona.</li></ul>                                                                                                                                                                                                                                                                                                                                                         |                                            |

## Licenses & Certifications

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**Databricks Machine Learning Engineer Associate**

June 2026

## Honors & Awards

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**Best Academic Record, Master's Degree in Advanced Mathematics**

July 2023

**Extraordinary Award, Bachelor's Degree in Mathematics**

July 2022

**Honorable Mention, Spanish Baccalaureate**

June 2018

## Technologies

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**Programming:** C, C++, R, Python, PySpark, Spark, SQL.

**Microsoft Azure Ecosystem:** Data Factory, Databricks, Azure Storage, Microsoft SQL Server & Azure Cosmos DB.

**Vector DB:** PostgreSQL.

**Version Control & Repositories:** Git, GitHub, Bitbucket.

**Machine Learning & AI Platforms:** Hugging Face.

**IDEs:** VS Code, Cursor, GitHub Copilot.

**Data Engineering Tools:** Docker, Kubernetes, Lens, Azure DevOps, Kibana (Elastic), RabbitMQ, FastAPI.

**Project Management & Agile:** Scrum, Agile Methodologies, Jira, Atlassian Ecosystem, Notion, Kanban.

## Projects & Expertise Areas

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**ML-Based Anomaly Detection & MLOps Implementation:** Developed a data-driven solution to proactively detect anomalous orders by combining machine learning, business rules, and scalable data pipelines. The system enables early identification of risks and prioritization of high-impact cases, improving operational efficiency and decision-making across the order-to-cash process.

- Developed an anomaly detection system leveraging supervised demand forecasting models to estimate expected order behavior at a customer-SKU-day level and identify statistically significant deviations.
- Built an end-to-end data pipeline integrating historical orders, contextual features, and feature engineering to support scalable outlier detection and classification of anomalous patterns.
- Established key MLOps practices such as model retraining cycles, champion-challenger evaluation, and CI/CD pipelines, enabling continuous improvement and robust performance in production.

**Search AI Engine:** Designed and deployed an *AI-powered search engine* capable of finding Points of Sale with photos and reviews most similar to a given image or short text input.

- Applied CLIP models (e.g., *SigLIP*, *Jina*) to generate embeddings for images and reviews.
- Utilized *GPUs* to accelerate embedding computation and *vector databases* (*PostgreSQL + HNSW indexing*) for efficient similarity search.
- Deployed a *production-ready API on Kubernetes* to integrate the AI search engine with the company's main platform, making it accessible to users.
- Developed an internal *Streamlit-based frontend* for the Data Science team, also used for client demos.

**Web Scraping:** Designed and maintained robust scrapers to extract information from multiple websites using Python libraries such as *BeautifulSoup* and *Selenium*. Ensured scalability, reliability, and adaptability to website changes.

**Data Pipelines & ETL:** Designed and implemented complex *ETL pipelines* to process large volumes of raw data into clean and structured datasets. Leveraged tools such as *Azure Storage*, *Azure Data Factory*, *Databricks*, and *Azure Cosmos DB* to orchestrate and optimize workflows.

**Client Ad-hoc Projects (CPG sector):** Developed tailored analytics solutions for international clients in the *Consumer Packaged Goods (CPG)* industry. Built ML models (*Linear/Logistic Regression*, *Naive Bayes*, *Decision Trees*, *XGBoost*, *Neu-*

ral Networks, Clustering, etc.) to identify Points of Sale (PoS) with the highest potential or specific characteristics, supporting commercial strategy.

## Publications

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**Quadratic differential systems with a WF1 and a finite saddle-node**

August 2023

*International Journal of Bifurcation and Chaos*. Joan C. Artés & **Carles Trullàs**

## Languages

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**Catalan and Spanish:** Native proficiency.

**English:** Full professional proficiency.